Aditya Samalla

(480) 758-7754 · aditya.samalla@asu.edu · adityaron.github.io · linkedin.com/in/adityasamalla

Summary

Engineering leader and Principal ML Engineer with 10+ years building production AI at scale. I founded and lead an LLM + RAG product line built on Claude, am a founding engineer and primary architect of the ML engine behind a flagship cloud-security product, and built the evaluation and red-teaming practice that keeps it safe and reliable. I turn frontier-model capabilities into measured outcomes — fewer false positives, multi-million-dollar savings, and revenue — and lead the teams that ship them.

Core Competencies

- LLMs & Small Language Models, RAG, Agentic / MCP
- LLM Evaluation, Red-teaming & AI Safety
- Transfer Learning, Deep Learning, MLOps
- Anomaly & Behavior Detection, Graph ML
- Production ML Systems & Data Platforms
- Engineering Management, Roadmaps, Mentoring
- **Languages:** Python, SQL, Scala, Go, R
- **AI/ML:** Claude (Bedrock), Azure OpenAI, LangChain, PyTorch, TensorFlow
- **Data:** Spark, Apache Iceberg/Lakehouse, Snowflake, Postgres, Neo4j, Airflow
- **Infra:** Kubernetes, Docker, AWS/GCP/Azure, Argo CD, Grafana

Work Experience

Fortinet (formerly Lacework) — **Aug 2022 – Present**

Software Engineering Manager / Principal Machine Learning Engineer — *Sunnyvale, California*

Promoted Senior → Staff Machine Learning Engineer → current role (3 levels, 2022–2026)

- **Founded and lead the AI Assist product line** — an LLM + RAG system (built on Claude via Amazon Bedrock) that turns complex cloud-security alerts into prioritized incident reports and alert-grounded remediation, exposed through an alert-context API for **MCP / agentic workflows**; set the roadmap at launch and shipped every milestone, driving **30% higher analyst engagement**.
- Built the **LLMOps and AI-safety practice**: an **LLM-as-judge evaluation harness**, **adversarial red-teaming (5,000+ prompts)**, production SLOs (p95 latency, error rate), and per-request token/cost observability.
- **Founding engineer and primary architect of the correlation engine** behind Composite Alerts — correlating **400+ detections** across **140+ MITRE ATT&CK techniques** and cloud/host/container/ML-anomaly signals into single high-fidelity incidents, **reducing false positives 40%**.
- Designed an **ephemeral graph-database microservice** within a Kubernetes multi-container framework to surface attack paths in potential breaches — **improving recall 15%, cutting latency 10×, and reducing data-warehouse cost $1M/year**.
- **Designed and led** the migration of the engine **off Snowflake onto an open Spark + Apache Iceberg lakehouse** — authoring the architecture (a producer/consumer split with single-scan grouped caching) and validating the production cutover — **cutting $1.5M+ in annualized cloud cost** (−53% warehouse, −57% compute).
- Built the **human-in-the-loop alert-labeling tool** and weak-supervision pipeline feeding four **constrained logistic-regression** alert-scoring models — precision gains that contributed to a **$1.2M enterprise deal**.
- Own a fleet of **10+ ML detectors** (GMM, Isolation Forest, ARIMA, CNN, NER+GBM) plus two **transfer-learned SLM detectors** (command-line, hostname-injection) — **+10% detection coverage**.
- Engineered an automated pipeline for **continuous evaluation** of detection effectiveness — precision/recall and ML metrics per detection to surface regressions early; integrated runtime vulnerable-activity (RiskWatch) signals into the correlation graph and shipped a customer-facing detection catalog (400+ detections).
- Manage and mentor a team across detection, GenAI, and platform; own the cross-team product-delivery roadmap, lead org-wide product launches, and set technical direction and code-review standards across the team's services.

Salesforce — Jun 2021 – Aug 2022

Senior Data Scientist — *San Francisco, California*

- Built and deployed supervised models (AWS SageMaker) forecasting customer subscription renewals for Tableau Online, **reducing churn 1.5%** at product scale.
- Applied topic modeling to autonomously group unstructured community ideas and stakeholder feedback for feature ranking, **increasing enhancement velocity 10%**.
- Designed supervised models using **SHAP** analysis to assess and improve Tableau visualization performance and user experience.
- Built data pipelines and models ingesting public census data for the **Tableau Data Equity Hub** — Tableau Foundation's data-for-good initiative helping organizations use data to advance equity and social justice.

Hewlett Packard Enterprise — Jun 2017 – Jun 2021

Senior Staff Data Scientist — *San Jose, California*

- Built machine learning for **HPE InfoSight**, HPE's AIOps platform — including a **workload anomaly detection** engine over storage-performance metrics with visual performance timelines.
- Engineered a storage **recommendation engine** that diagnosed performance issues and recommended optimizations — **automatically resolving 86% of customer cases**.
- Built **cross-stack analytics for virtual environments** — attributing latency across host, storage, and network — **resolving 54% of issues beyond the storage layer**; created an "Impact Score" to quantify severity.
- Applied **dimensionality reduction and unsupervised learning** to high-dimensional telemetry to surface natural variability and cluster behavior, and built diagnostic signatures through cross-team collaboration with support, performance engineering, and product.
- Developed **capacity forecasting and sizing** models for performance and data-driven sales — **driving 15% revenue and 10% sales-efficiency gains**; architected lead-generation integration enabling **$3M+ in opportunities**.

Nimble Storage — Sep 2016 – May 2017

Data Scientist — *San Jose, California*

- Built statistical and machine-learning models to proactively predict storage issues and performance bottlenecks — part of Nimble's predictive-analytics work that became HPE InfoSight.
- Designed executive data-visualization dashboards (Tableau / R Shiny) and built Flask APIs feeding downstream systems — driving KPI tracking and data-driven decisions across marketing, sales, and pricing with cross-functional partners.
- Ran exploratory data analysis on operational, sales, and support metrics and automated the supporting data pipelines, **improving support satisfaction 12.5%**.

Accomplishments

- **Patent (Issued, USPTO):** US 12,627,687 — Generative AI-Based Multi-stage Composite Event Processing — co-inventor.
- **Patent (Granted):** US 9,065,952 B2 — robust determination of the significance of performance issues in IT environments. **Patent (Pending):** ML-based detection of compromised AWS credentials. **Patent (Filing Approved):** LLM-based triage of cloud alerts using historical context and customer-provided guidance.
- **Publication / blog:** "FortiCNAPP AI Assist: Accelerating the Investigation and Remediation of Cloud Threats" (Fortinet Community, 2026); "Performance Analysis of Noise Cancellation in Speech Signals Using LMS, FT-LMS and RLS Algorithms."
- Member, **Beta Gamma Sigma** — International Honor Society (highest scholastic recognition). Presented ML features at HPE Techcon, Discover, and storage conferences.

Education

W. P. Carey School of Business, Arizona State University — *M.S. in Analytics, GPA 4.00* — **2016**

Jawaharlal Nehru Technological University — *B.Tech, Electronics & Communication Engineering* — **2011**